k) **Span:** This is the clear horizontal distance between the two supports.

l) **Rise:** This is the vertical distance between the highest point of intrados and springing line.

m) **Depth of arch**: This is the perpendicular distance between the extrados and intrados.

n) **Haunch of an arch**: Haunch is the lower part of the arch between crown and skewback.

7.1.5 Classification of Arches

A) Classification according to shape.

According to this classification, arches may be of the following types.

1) Flat arch
2) Segmental arch
3) Semi-circular arch
4) Horse shoe arch
5) Pointed arch or gothic arch
6) Relieving arch
7) Circular arch
8) Inverted arch

B) Classification based on materials of construction.

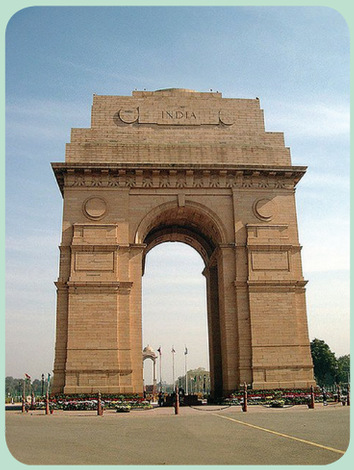
1) Stone arches
2) Brick arches
3) Concrete arches

Flat Arch

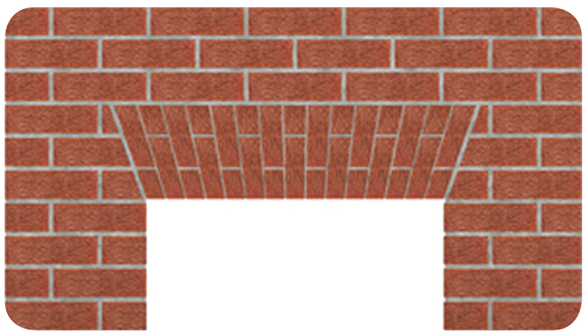
A flat arch usually the angle formed by skew backs is 60° with horizontal, thus forming an equilateral triangle with intrados at the base. The intrados is apparently flat, but it is given a slight rise of camber of about 10mm to 15mm per meter width of opening to allow for small settlements.



India gate, New Delhi, India is one of the most popular arch in the world. This monuments dedicated to Indian soldiers who died in World War 1 and the Afghan wars.



However, the extrados is kept horizontal and flat. Flat arches are used only for light loads and for spans up to 1.5m.



Segmental Arch

This is common type of arch used for buildings. The centre point of these arches lies below the springing line. It is one of the strongest arch because, it is able to resist thrust. Segmental arches are most commonly used for residential

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